

Initial Data Summary Report

Dataset Overview

The Customer Personality Analysis dataset contains demographic information, customer purchasing behavior, campaign responses, and customer engagement data. The dataset is used to analyze customer segments and understand purchasing patterns for business decision-making and machine learning applications.

Number of Variables

The dataset contains **29 variables (columns)**.

Number of Records

The dataset contains **2240 customer records (rows)**.

Numerical Variables

The numerical variables include customer ID, Year_Birth, Income, Kidhome, Teenhome, Recency, product spending variables, purchase counts, web visits, campaign responses, complaints, and total purchases.

Categorical Variables

The categorical variables include:

- Education
- Marital_Status
- Dt_Customer

Missing Values

The **Income** column contains missing values. Most other variables do not contain missing data.

Duplicate Records

No complete duplicate rows were found in the dataset. Customer IDs are unique.

Major Data Quality Issues

The following issues were identified:

- Missing values in Income.
- Some unusual values in Year_Birth that may represent unrealistic customer ages.
- Marital_Status contains categories that can be standardized (such as "Alone", "YOLO", "Absurd").
- Dt_Customer should be converted from text to DateTime format.
- Outliers exist in Income and product spending variables.

Conclusion

Overall, the dataset is well structured and suitable for machine learning after handling missing values, correcting data types, treating outliers, and standardizing categorical values.